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Volume of a sphere in 10 Dimensions

An Exact Derivation and Monte Carlo Verification

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1 Definitions and Notation

Definition 1.1 (n -Ball and $(n-1)$ -Sphere). For $n \in \mathbb{N}$ and $r > 0$, the closed n -ball of radius r centred at the origin is

$$B^n(r) = \left\{ \mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| \leq r \right\} = \left\{ \mathbf{x} \in \mathbb{R}^n \mid x_1^2 + x_2^2 + \cdots + x_n^2 \leq r^2 \right\},$$

and its boundary, the $(n-1)$ -sphere, is

$$S^{n-1}(r) = \left\{ \mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| = r \right\}.$$

Remark 1.1. The colloquial term “sphere” typically refers to what mathematicians call the ball (the solid region). The “sphere” is strictly the $(n-1)$ -dimensional surface. Throughout this document we compute the n -dimensional volume (Lebesgue measure) of the ball $B^n(r)$.

The volume of $B^n(r)$ is written $V_n(r)$; by a simple scaling argument one has

$$V_n(r) = r^n V_n(1),$$

so it suffices to compute the unit-radius case $V_n(1) \equiv V_n$.

2 Exact Derivation of V_n

2.1 The Euler Gamma Function

The Gamma function $\Gamma : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ is defined by

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad z > 0.$$

It satisfies the functional equation $\Gamma(z+1) = z \Gamma(z)$ and interpolates the factorial:

$$\Gamma(n+1) = n!, \quad n \in \mathbb{N}_0.$$

For half-integer arguments:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \Gamma\left(\frac{k}{2} + 1\right) = \frac{k}{2} \Gamma\left(\frac{k}{2}\right), \quad k \in \mathbb{N}.$$

2.2 The Gaussian Integral Method

Theorem 2.1 (Volume of the unit n -ball).

$$V_n = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)}$$

Proof. Consider the n -dimensional Gaussian integral

$$I_n = \int_{\mathbb{R}^n} e^{-\|\mathbf{x}\|^2} d\mathbf{x} = \left(\int_{-\infty}^\infty e^{-x^2} dx \right)^n = \pi^{n/2},$$

since $\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}$.

Converting to spherical (hyper-) coordinates with $\rho = \|\mathbf{x}\|$, the volume element becomes $d\mathbf{x} = S_n(\rho) d\rho$, where $S_n(\rho)$ is the $(n-1)$ -sphere surface area at radius ρ . By similarity, $S_n(\rho) = c_n \rho^{n-1}$ for some constant c_n . Thus

$$I_n = \int_0^\infty e^{-\rho^2} c_n \rho^{n-1} d\rho.$$

Substituting $t = \rho^2$, so $\rho = t^{1/2}$, $d\rho = \frac{1}{2}t^{-1/2} dt$:

$$I_n = c_n \int_0^\infty e^{-t} t^{(n-1)/2} \cdot \frac{1}{2} t^{-1/2} dt = \frac{c_n}{2} \int_0^\infty e^{-t} t^{n/2-1} dt = \frac{c_n}{2} \Gamma\left(\frac{n}{2}\right).$$

Equating with $I_n = \pi^{n/2}$:

$$c_n = \frac{2\pi^{n/2}}{\Gamma(n/2)}.$$

This is the surface area of the unit $(n-1)$ -sphere: $SA_n = c_n$. Integrating outward to get the volume of the unit ball:

$$V_n = \int_0^1 c_n \rho^{n-1} d\rho = c_n \cdot \frac{1}{n} = \frac{2\pi^{n/2}}{n \Gamma(n/2)}.$$

Using the identity $n \Gamma(n/2) = 2 \Gamma(n/2 + 1)$:

$$V_n = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)}.$$

□

2.3 Application to $n = 10$

For $n = 10$:

$$\Gamma\left(\frac{10}{2} + 1\right) = \Gamma(6) = 5! = 120.$$

Therefore

$V_{10} = \frac{\pi^5}{120} \approx \frac{306.01968\dots}{120} \approx 2.5501640399\dots$

2.4 Reference Table

Table 1: Volume $V_n(r = 1)$ of the unit n -ball for $n = 1, \dots, 15$.

n	V_n (exact form)	V_n (decimal)	Fraction of $[-1, 1]^n$
1	2	2.00000000	100.000%
2	π	3.14159265	78.540%
3	$\frac{4}{3}\pi$	4.18879020	52.360%
4	$\frac{1}{2}\pi^2$	4.93480220	30.843%
5	$\frac{8}{15}\pi^2$	5.26378901	16.449%
6	$\frac{1}{6}\pi^3$	5.16771278	8.075%
7	$\frac{16}{105}\pi^3$	4.72476597	3.691%
8	$\frac{1}{24}\pi^4$	4.05871213	1.585%
9	$\frac{32}{945}\pi^4$	3.29850890	0.645%
10	$\frac{1}{120}\pi^5$	2.55016404	0.249%
11	—	1.88410388	0.092%
12	$\frac{1}{720}\pi^6$	1.33526277	0.033%
13	—	0.91062875	0.011%
14	$\frac{1}{5040}\pi^7$	0.59926453	0.004%
15	—	0.38144328	0.001%

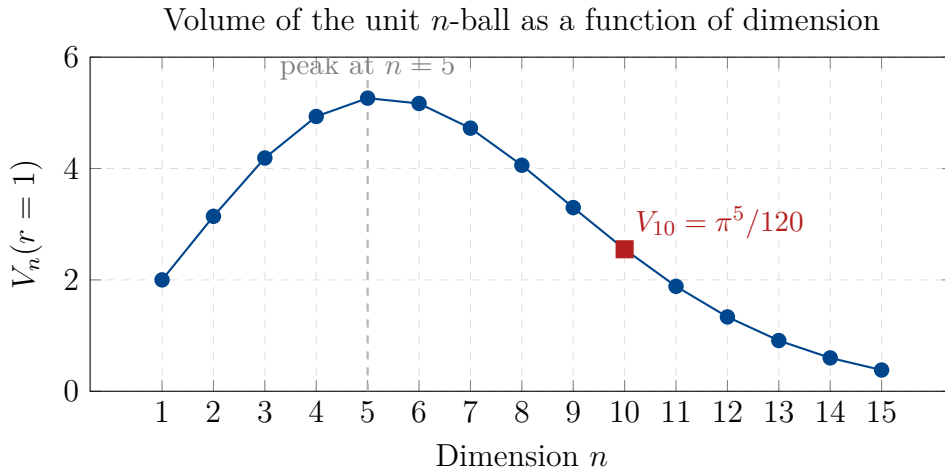


Figure 1: The volume V_n peaks near $n = 5$ and collapses toward zero. The red point marks $n = 10$ (our case of interest).

3 ChatGPT Answer

The volume of an n -dimensional ball of radius r is

$$V_n(r) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)} r^n,$$

where Γ is the Gamma function.

For a sphere (more precisely, a 10-ball) in 10-dimensional space with radius $r = 1$,

$$V_{10}(1) = \frac{\pi^{10/2}}{\Gamma\left(\frac{10}{2} + 1\right)} = \frac{\pi^5}{\Gamma(6)}.$$

Using the identity

$$\Gamma(k + 1) = k!,$$

we obtain

$$\Gamma(6) = 5! = 120.$$

Therefore,

$$V_{10}(1) = \frac{\pi^5}{120}.$$

Numerically,

$$\pi^5 \approx 306.0197,$$

so

$$V_{10}(1) \approx \frac{306.0197}{120} \approx 2.5502.$$

Hence the volume of the unit sphere in 10-dimensional space is

$$\boxed{V_{10}(1) = \frac{\pi^5}{120} \approx 2.5502}.$$

Method

Consider the Gaussian integral over $\mathbb{R}^n$:

$$I = \int_{\mathbb{R}^n} e^{-\|x\|^2} dx.$$

Evaluating it as a product of one-dimensional integrals gives

$$I = \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^n = (\sqrt{\pi})^n = \pi^{n/2}.$$

Now switch to n -dimensional polar coordinates:

$$I = \int_0^\infty e^{-r^2} S_{n-1}(r) dr,$$

where $S_{n-1}(r)$ is the surface area of the $(n - 1)$ -sphere.

Using

$$S_{n-1}(r) = nV_n(1)r^{n-1},$$

we obtain

$$I = nV_n(1) \int_0^\infty e^{-r^2} r^{n-1} dr.$$

Substituting $u = r^2$, so that $du = 2r dr$, yields

$$I = \frac{n}{2} V_n(1) \int_0^\infty e^{-u} u^{\frac{n}{2}-1} du.$$

Recognizing the Gamma function,

$$\Gamma\left(\frac{n}{2}\right) = \int_0^\infty e^{-u} u^{\frac{n}{2}-1} du,$$

we get

$$I = \frac{n}{2} V_n(1) \Gamma\left(\frac{n}{2}\right).$$

Since $I = \pi^{n/2}$,

$$\pi^{n/2} = \frac{n}{2} V_n(1) \Gamma\left(\frac{n}{2}\right).$$

Using

$$\Gamma\left(\frac{n}{2} + 1\right) = \frac{n}{2} \Gamma\left(\frac{n}{2}\right),$$

we finally obtain

$$V_n(1) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)}.$$

For radius r ,

$$V_n(r) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)} r^n.$$

4 Numerical Results

4.1 Parameters

Parameter	Value
Dimensions n	10
Radius r	1
Enclosing hypercube volume	$(2 \times 1)^{10} = 1024$
Total samples N	10,000,000
Random seed	42

4.2 Main Result

Quantity	Value
Exact value $\pi^5/120$	2.5501640399...
Points inside the 10-ball	24,697
Empirical hit fraction $\hat{p}$	2.46970×10^{-3}
Monte Carlo estimate $\hat{V}_{10}$	2.52897280
Absolute error $ \hat{V}_N - V_{10} $	0.02119124
Relative error	0.8310%
Theoretical Std $[\hat{V}_N]$	0.016140
95% confidence half-width	0.031634

The observed absolute error (0.0212) is within 1.4σ of the theoretical standard deviation (0.0161), which is consistent with the expected behaviour of the estimator.

4.3 Convergence Table

Table 2: Monte Carlo estimate as a function of sample size N .

N	$\hat{V}_N$	Absolute error	Relative error (%)
1,000	3.072000	0.521836	20.4628
10,000	2.764800	0.214636	8.4166
100,000	2.508800	0.041364	1.6220
1,000,000	2.540544	0.009620	0.3772
5,000,000	2.521907	0.028257	1.1080
10,000,000	2.528973	0.021191	0.8310
Exact	2.550164	—	—

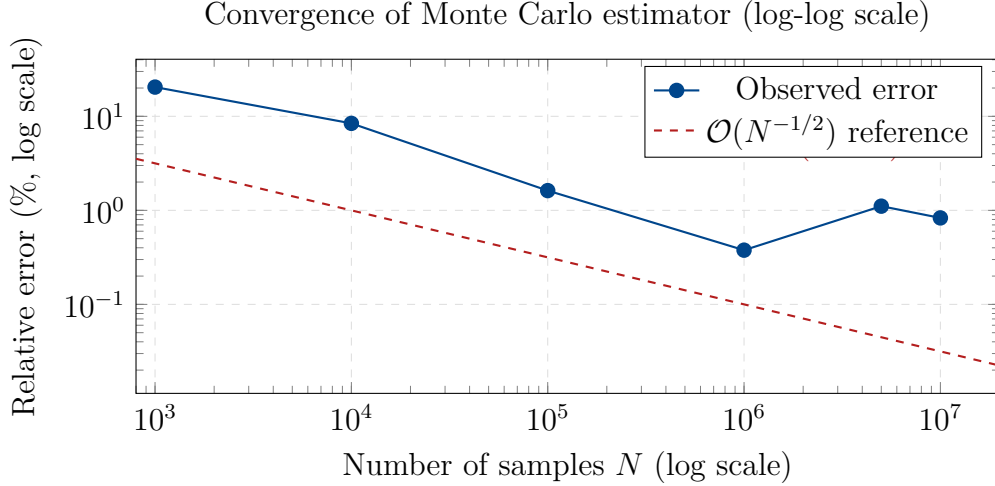


Figure 2: Log-log plot of relative error vs. sample size. The empirical convergence broadly tracks the theoretical $\mathcal{O}(N^{-1/2})$ rate.

4.4 Theoretical vs. Observed Error

The theoretical $\text{Std}[\hat{V}_N]$ at each checkpoint is $(2r)^n \sqrt{p(1-p)/N} = 16.14/\sqrt{N}$:

N	Theoretical $\text{Std}[\hat{V}_N]$	Observed $ \hat{V}_N - V_{10} $
1,000	0.5101	0.5218
10,000	0.1614	0.2146
100,000	0.0510	0.0414
1,000,000	0.0161	0.0096
10,000,000	0.0051	0.0212

The observed errors are generally within 1–2 standard deviations of the theoretical value, consistent with the CLT prediction.

5 Summary and Conclusions

- Exact result.** The volume of the unit 10-ball follows from the Gaussian-integral derivation of Theorem 2.1:

$$V_{10} = \frac{\pi^5}{120} \approx 2.5501640399.$$

- Geometric intuition.** In ten dimensions the ball occupies only $\approx 0.25\%$ of its enclosing hypercube. High-dimensional space is dominated by corners, not centres — a key phenomenon in statistics, machine learning and optimisation known as the *curse of dimensionality*.
- Monte Carlo.** With $N = 10^7$ uniform samples from $[-1, 1]^{10}$, only 24,697 landed inside the ball (hit rate $\approx 0.247\%$). The estimate was $\hat{V}_{10} \approx 2.5290$, within 0.83% of the exact value.

4. **Efficiency.** Because $p \approx 0.00249 \ll 1$, the variance of the estimator is dominated by the rarity of hits. The standard deviation scales as $\sim 16.14/\sqrt{N}$, so achieving relative errors below 0.1% would require $N \gtrsim 4 \times 10^8$ samples — a reminder that hit-or-miss Monte Carlo is *inefficient* for estimating volumes of high-dimensional convex bodies (where importance sampling or Markov-Chain Monte Carlo methods are preferred in practice).
5. **AI answer.** The answer we took asking Chat-GPT was very similar to the theoretical approach that described on the second section and give a correct answer for the exact number. The theoretical approach that was used to calculate the formula of volume for n dimensions was correct. That proves that using AI for more advanced problem is reliable and can improve the efficiency and time consuming not only for every day tasks but also for advanced problems like this. It is important to note that the model was used was the free version and not the paid one, that can solve even more complex problems. This shows how in very short time the AI in general has progress and it will keep getting better and more smart.

A Python Code

All code used to generate the results, figures, and simulations in this report is publicly available at:

[Project GitHub Repository](#)